

Gel electrophoresis is a widely used technique that separates charged macromolecules, such as negatively charged DNA and RNA, according to their molecular size. An electrophoretic system operates as an electrolytic cell in which a direct electric current is applied to two metal electrodes immersed in a conducting electrolyte solution. The force from the resulting electric field causes the charged macromolecules to migrate through a porous agarose gel at a rate that is inversely related to their size. A schematic of an electrophoretic system is shown in Figure 1.

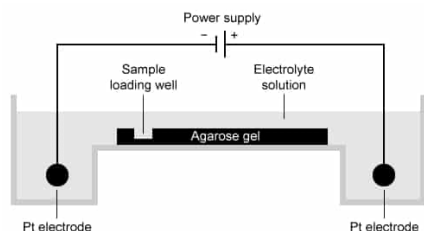
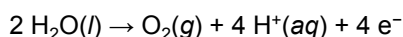
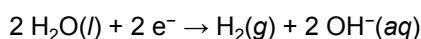


Figure 1 Agarose gel electrophoretic system

During electrophoresis, electron transfer reactions occur at the interface where the surface of each electrode contacts the electrolyte solution. Typically, the electrochemical reaction occurring is the electrolysis of water, generating $\text{H}_2(g)$ and $\text{O}_2(g)$ bubbles at the surface of the electrodes according to Reactions 1 and 2.



Reaction 1



Reaction 2

In an experiment, a researcher uses agarose gel electrophoresis to analyze the size distribution of different DNA samples. The researcher prepares a $5.0 \times 10^{-3} \text{ M}$ lithium acetate solution, which is then poured into the electrophoretic tank. After loading the samples into the agarose gel, a current is applied until the DNA fragments are separated (seen as individual bands in the gel). The agarose gel after electrophoresis is shown in Figure 2.

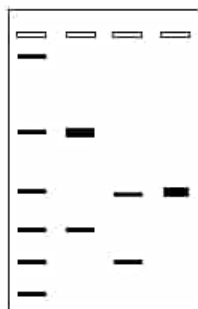


Figure 2 Agarose gel after electrophoresis using a $5.0 \times 10^{-3} \text{ M}$ lithium acetate solution (Note: The DNA fragments are represented as black bands.)

How many grams of lithium acetate were mixed in 500. mL of water to prepare the $5.0 \times 10^{-3} \text{ M}$ lithium acetate solution? (Note: The molar mass of lithium acetate is 66 g/mol.)

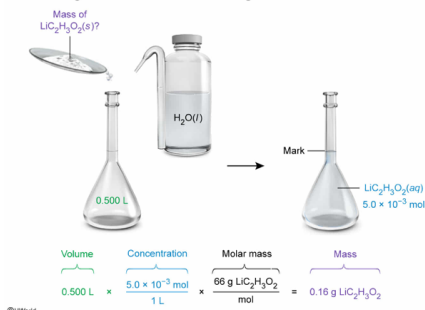
- ☐ A. $3.8 \times 10^{-5} \text{ g}$ [4%]
- ☐ B. $2.5 \times 10^{-3} \text{ g}$ [8%]
- ☒ C. $1.6 \times 10^{-1} \text{ g}$ [80%]
- ☐ D. $1.6 \times 10^2 \text{ g}$ [6%]

Correct

79% answered correctly

Explanation:

Determining the mass of lithium acetate given volume and concentration



The concentration of a solution describes the quantity of solute dissolved per unit volume of solution. The **molar concentration** (ie, **molarity M**) of a solution is defined as the **moles** of dissolved solute per **liter** of solution:

$$\text{Molarity} = \frac{\text{moles of solute}}{\text{solution volume (in liters)}}$$

As such, the moles of $\text{LiC}_2\text{H}_3\text{O}_2$ needed to prepare the given solution can be found by multiplying the given volume (in liters) by the specified concentration.

$$500. \text{ mL} \times \frac{1 \text{ L}}{1,000 \text{ mL}} = 0.500 \text{ L}$$

$$0.500 \text{ L} \times \frac{5.0 \times 10^{-3} \text{ mol LiC}_2\text{H}_3\text{O}_2}{1 \text{ L}} = 2.5 \times 10^{-3} \text{ mol LiC}_2\text{H}_3\text{O}_2$$

Multiplying the moles of $\text{LiC}_2\text{H}_3\text{O}_2$ by the **molar mass** of $\text{LiC}_2\text{H}_3\text{O}_2$ yields the mass of $\text{LiC}_2\text{H}_3\text{O}_2(\text{s})$ in grams.

$$2.5 \times 10^{-3} \text{ mol LiC}_2\text{H}_3\text{O}_2 \times \frac{66 \text{ g LiC}_2\text{H}_3\text{O}_2}{\text{mol LiC}_2\text{H}_3\text{O}_2} = 0.16 \text{ g LiC}_2\text{H}_3\text{O}_2$$

Therefore, 0.16 g (ie, $1.6 \times 10^{-1} \text{ g}$) of $\text{LiC}_2\text{H}_3\text{O}_2(\text{s})$ are needed to prepare 500. mL of $5.0 \times 10^{-3} \text{ M LiC}_2\text{H}_3\text{O}_2(\text{aq})$.

(Choice A) An amount of $3.8 \times 10^{-5} \text{ g}$ (with incorrect units) is obtained if the moles of $\text{LiC}_2\text{H}_3\text{O}_2$ are *divided* by its molar mass instead of *multiplied* by the molar mass.

(Choice B) A value of 2.5×10^{-3} is numerically equal to the *moles* of $\text{LiC}_2\text{H}_3\text{O}_2$, *not* the mass.

(Choice D) If the volume is not converted from milliliters to liters, a quantity of $1.6 \times 10^2 \text{ g}$ (with incorrect units) is obtained.

Educational objective:

The molarity of a solution is defined as the moles of dissolved solute per liter of solution. The amount of solute in moles needed to prepare a solution can be found by multiplying the solution volume by the desired molarity. The mass of the solute can then be calculated by multiplying the moles of the solute by its molar mass.

Time spent: 0 sec
Subject:
General Chemistry

QID: 403374
System:
Solutions and Electrochemistry

Gel electrophoresis is a widely used technique that separates charged macromolecules, such as negatively charged DNA and RNA, according to their molecular size. An electrophoretic system operates as an electrolytic cell in which a direct electric current is applied to two metal electrodes immersed in a conducting electrolyte solution. The force from the resulting electric field causes the charged macromolecules to migrate through a porous agarose gel at a rate that is inversely related to their size. A schematic of an electrophoretic system is shown in Figure 1.

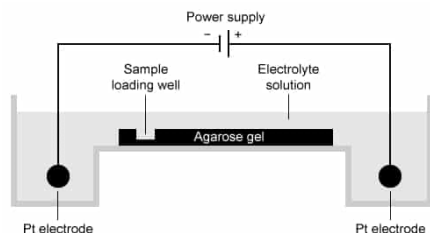
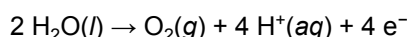
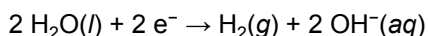


Figure 1 Agarose gel electrophoretic system

During electrophoresis, electron transfer reactions occur at the interface where the surface of each electrode contacts the electrolyte solution. Typically, the electrochemical reaction occurring is the electrolysis of water, generating $\text{H}_2(g)$ and $\text{O}_2(g)$ bubbles at the surface of the electrodes according to Reactions 1 and 2.



Reaction 1



Reaction 2

In an experiment, a researcher uses agarose gel electrophoresis to analyze the size distribution of different DNA samples. The researcher prepares a $5.0 \times 10^{-3} \text{ M}$ lithium acetate solution, which is then poured into the electrophoretic tank. After loading the samples into the agarose gel, a current is applied until the DNA fragments are separated (seen as individual bands in the gel). The agarose gel after electrophoresis is shown in Figure 2.

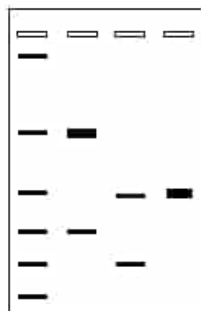


Figure 2 Agarose gel after electrophoresis using a $5.0 \times 10^{-3} \text{ M}$ lithium acetate solution (Note: The DNA fragments are represented as black bands.)

When the power source is connected to the platinum electrodes (Figure 1), which of the following statements are true?

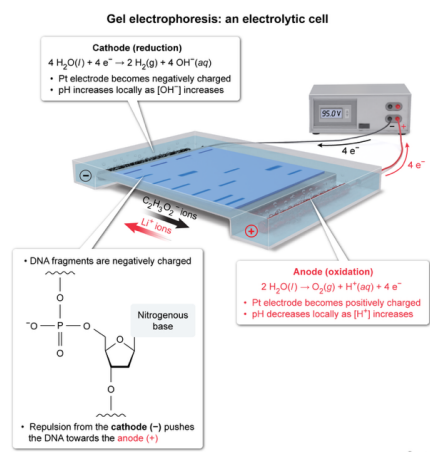
- I. DNA migrates toward the anode.
- II. pH increases locally at the cathode.
- III. Acetate ions move toward the cathode.
- IV. $\text{O}_2(g)$ evolves at the anode.

- ☐ A. I and II only [13%]
- ☐ B. II and III only [14%]
- ✓ ☒ C. I, II, and IV only [47%]
- ☐ D. II, III, and IV only [25%]

Correct

46% answered correctly

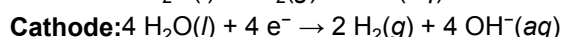
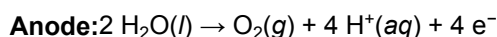
Explanation:



In an **electrolytic cell**, two metal electrodes immersed in a **conducting medium** are connected to an external power source. The external power source removes electrons from the electrode called the **anode** (positively charged), where **oxidation** takes place, and transfers them to the electrode called the **cathode** (negatively charged), where **reduction** takes place. The electron flow (ie, **electric current**) generated by the external source is then used to drive **nonspontaneous** processes.

In the passage, the **gel electrophoretic** system operates as an electrolytic cell. Connecting the platinum electrodes (immersed in an aqueous $\text{LiC}_2\text{H}_3\text{O}_2$ solution) to an external power source generates a current that causes the DNA fragments to migrate through the gel matrix. As the electrophoretic system operates, the negatively charged phosphate groups of the DNA fragments are **repelled** by the negatively charged cathode and attracted by the positively charged anode. Consequently, the fragments migrate through the gel matrix toward the positively charged anode (**Number I**).

In addition, at the interface where the platinum electrodes contact the aqueous solution, water is oxidized at the anode and reduced at the cathode (ie, water electrolysis):



Because $\text{OH}^-(aq)$ ions are formed through water oxidation, the **pH** of the solution increases locally at the cathode as the concentration of $\text{OH}^-(aq)$ ions increases (**Number II**). Simultaneously, water oxidation produces $\text{O}_2(g)$ at the anode (**Number IV**).

(**Number III**) Acetate ions ($\text{C}_2\text{H}_3\text{O}_2^-$) are *negatively* charged and move away from (not toward) the cathode in the direction of the *anode*.

Educational objective:

An electrolytic cell uses an electric current generated by an external source to drive a nonspontaneous process. Oxidation occurs at the positively charged anode and reduction occurs at the negatively charged cathode. In an electrochemical cell, electrolyte ions flow toward the oppositely charged electrode.

Time spent: 0 sec
Subject:
General Chemistry

QID: 403375
System:
Solutions and Electrochemistry

Gel electrophoresis is a widely used technique that separates charged macromolecules, such as negatively charged DNA and RNA, according to their molecular size. An electrophoretic system operates as an electrolytic cell in which a direct electric current is applied to two metal electrodes immersed in a conducting electrolyte solution. The force from the resulting electric field causes the charged macromolecules to migrate through a porous agarose gel at a rate that is inversely related to their size. A schematic of an electrophoretic system is shown in Figure 1.

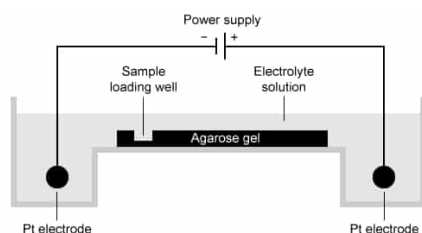
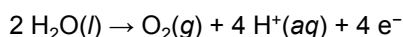
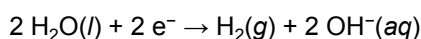


Figure 1 Agarose gel electrophoretic system

During electrophoresis, electron transfer reactions occur at the interface where the surface of each electrode contacts the electrolyte solution. Typically, the electrochemical reaction occurring is the electrolysis of water, generating $\text{H}_2(g)$ and $\text{O}_2(g)$ bubbles at the surface of the electrodes according to Reactions 1 and 2.



Reaction 1



Reaction 2

In an experiment, a researcher uses agarose gel electrophoresis to analyze the size distribution of different DNA samples. The researcher prepares a $5.0 \times 10^{-3} \text{ M}$ lithium acetate solution, which is then poured into the electrophoretic tank. After loading the samples into the agarose gel, a current is applied until the DNA fragments are separated (seen as individual bands in the gel). The agarose gel after electrophoresis is shown in Figure 2.

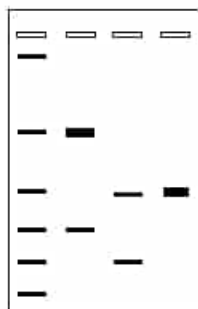


Figure 2 Agarose gel after electrophoresis using a $5.0 \times 10^{-3} \text{ M}$ lithium acetate solution (Note: The DNA fragments are represented as black bands.)

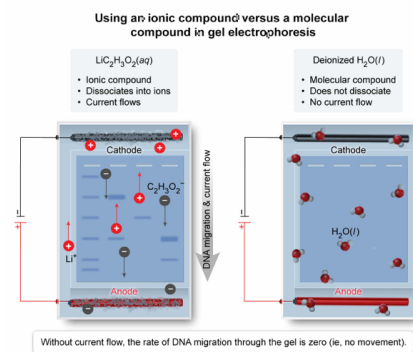
How will the rate of DNA migration through the gel be affected if the gel is run with deionized water rather than $\text{LiC}_2\text{H}_3\text{O}_2(aq)$?

- ☐ A. The rate of migration will remain the same. [6%]
- ✓ ☒ B. The rate of migration will decrease to zero. [74%]
- ☐ C. The rate of migration will increase. [12%]
- ☐ D. The rate of migration is unrelated to the electrolyte solution used. [7%]

Correct

73% answered correctly

Explanation:



Gel electrophoresis, acting as an **electrolytic cell**, uses an external power source to provide an **electric current** (ie, the movement of electrons) that travels from the **cathode** (where **reduction** occurs) to the **anode** (where oxidation occurs). Recall that by convention, the direction of the electric current is opposite the direction of the electron flow.

Within the electrophoresis tank, charge balance is maintained using a **conductive electrolytic solution**. Cations and anions (ie, charge-carrying electrolytes) present within the electrolytic solution migrate toward the **negatively charged** cathode and **positively charged** anode, respectively, resulting in current flow through the gel. Without current flow, negatively charged DNA cannot migrate through the gel.

The gel electrophoresis configuration described in the passage uses $\text{LiC}_2\text{H}_3\text{O}_2(aq)$ (ie, an **ionic salt solution**) as the conductive medium. When dissolved, ionic salts dissociate into ions (eg, Li^+ and $\text{C}_2\text{H}_3\text{O}_2^-$) that can **freely move** and assist the passage of electrons through the solution. Alternatively, molecular liquids such as deionized $\text{H}_2\text{O}(l)$ are nonelectrolytes because they do not dissociate into ions and do not convey electrons through the gel.

Therefore, replacing $\text{LiC}_2\text{H}_3\text{O}_2(aq)$ with deionized $\text{H}_2\text{O}(l)$ would prevent current flow, and the rate of DNA migration through the gel would be zero (ie, no movement through the gel).

(Choices A, C, and D) Current cannot flow through the electrophoresis gel without a conductive electrolytic solution to facilitate the migration and separation of DNA fragments.

Educational objective:

During gel electrophoresis, a conductive electrolytic solution maintains a charge balance at the anode and cathode and assists the passage of electrons through the gel and solution. Without an electrolyte, current cannot flow and the DNA will not migrate through the gel.

Time spent: 0 sec
Subject:
General Chemistry

QID: 403376
System:
Solutions and Electrochemistry

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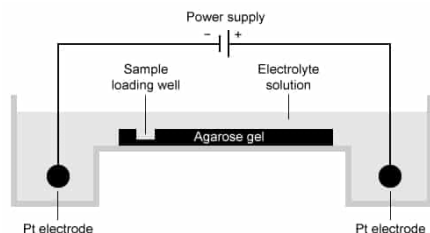
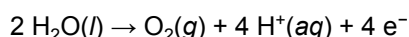
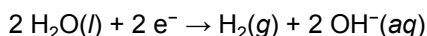


Figure 1 Agarose gel electrophoretic system

During electrophoresis, electron transfer reactions occur at the interface where the surface of each electrode contacts the electrolyte solution. Typically, the electrochemical reaction occurring is the electrolysis of water, generating $\text{H}_2(g)$ and $\text{O}_2(g)$ bubbles at the surface of the electrodes according to Reactions 1 and 2.



Reaction 1



Reaction 2

In an experiment, a researcher uses agarose gel electrophoresis to analyze the size distribution of different DNA samples. The researcher prepares a $5.0 \times 10^{-3} \text{ M}$ lithium acetate solution, which is then poured into the electrophoretic tank. After loading the samples into the agarose gel, a current is applied until the DNA fragments are separated (seen as individual bands in the gel). The agarose gel after electrophoresis is shown in Figure 2.

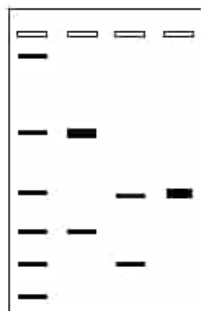


Figure 2 Agarose gel after electrophoresis using a $5.0 \times 10^{-3} \text{ M}$ lithium acetate solution (Note: The DNA fragments are represented as black bands.)

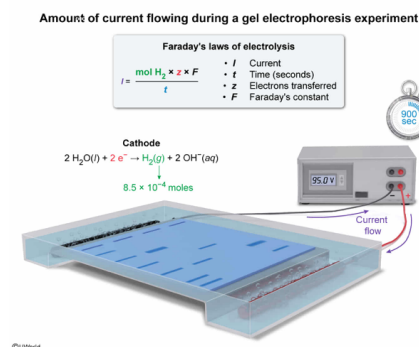
What expression gives the average amount of current (in amps) that flows through the agarose gel if 8.5×10^{-4} moles of $\text{H}_2(g)$ are produced in 900 seconds (15 minutes)? (Note: $F = 9.6 \times 10^4 \text{ C mol}^{-1}$)

- ✓ ☒ A. $(8.5 \times 10^{-4}) \times 2 \times (9.6 \times 10^4) / (9.0 \times 10^2)$ [53%]
☐ B. $(8.5 \times 10^{-4}) \times (9.6 \times 10^4) / (9.0 \times 10^2)$ [39%]
☐ C. $(8.5 \times 10^{-4}) \times 4 \times (9.6 \times 10^4) / 15$ [4%]
☐ D. $(8.5 \times 10^{-4}) \times 2 \times (9.6 \times 10^4)$ [2%]

Correct

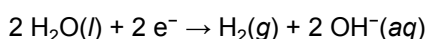
52% answered correctly

Explanation:



In an **electrolytic cell**, electric current supplied from an external source is used to drive a **nonspontaneous redox** reaction. This process, known as electrolysis, electrochemically **decomposes** (ie, consumes) a compound by **oxidizing** electron-rich species at the **anode** and **reducing** electron-poor species at the **cathode**.

During the electrolysis of water, electrical energy is used to decompose water into hydrogen gas and hydroxide ions at the cathode by means of reduction, according to Reaction 2.



The question states that 8.5×10^{-4} mol of $\text{H}_2(g)$ were produced after operating the electrophoretic system for 15 min (900 s). Based on the **stoichiometric coefficients** of the reduction reaction, 2 moles of electrons are transferred for each mole of $\text{H}_2(g)$ that is formed. As such, the total amount of electrons transferred is evaluated as:

$$8.5 \times 10^{-4} \text{ mol H}_2(g) \times \frac{2 \text{ mol e}^-}{1 \text{ mol H}_2(g)} = (8.5 \times 10^{-4}) \times 2 \text{ mol e}^-$$

Current is measured in amperes, where 1 ampere (A) equals 1 coulomb (C) of electric charge flowing through the gel per second. Moles of e^- can be converted to coulombs using **Faraday's constant**:

$$(8.5 \times 10^{-4}) \times 2 \text{ mol e}^- \times \frac{9.6 \times 10^4 \text{ C}}{1 \text{ mol e}^-} = (8.5 \times 10^{-4}) \times 2 \times (9.6 \times 10^4 \text{ C})$$

Dividing the above expression by 900 s yields the amount of charge flowing through the gel per second:

$$\frac{(8.5 \times 10^{-4}) \times 2 \times (9.6 \times 10^4 \text{ C})}{900 \text{ s}}$$

Therefore, the expression for current flowing through the gel can be written as

$$\left[\frac{(8.5 \times 10^{-4}) \times 2 \times (9.6 \times 10^4)}{9.0 \times 10^{-2}} \right] \text{ amp}$$

(Choice B) This expression neglects the fact that 2 moles of electrons are transferred per 1 mole of $\text{H}_2(g)$ formed at the cathode.

(Choice C) This expression is obtained by assuming that 4 moles of electrons are transferred for each mole of $\text{H}_2(g)$ formed; however, only 2 moles of electrons are transferred per mole of $\text{H}_2(g)$ produced. In addition, 1 ampere of current is defined as 1 coulomb of electric charge per *second*, not per minute.

(Choice D) This expression represents the total amount of charge in *coulombs* passing through the gel over a period of 15 min. However, *current* is the *flow* of electric charge in coulombs *per second*.

Educational objective:

An electrolytic cell uses an electric current from an external source to drive the nonspontaneous electrochemical decomposition of a compound. The average amount of current applied during an interval can be determined by multiplying the total number of electrons transferred in an electrochemical conversion by Faraday's constant and then dividing by the time (in seconds).

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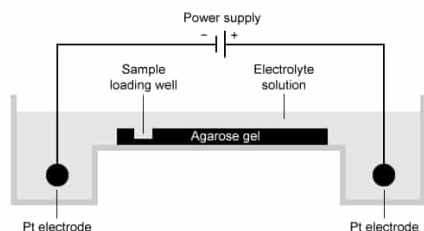
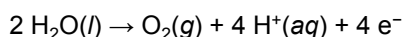
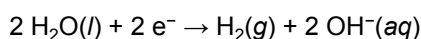


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Reaction 1



Reaction 2

In an experiment, a researcher uses agarose gel electrophoresis to analyze the size distribution of different DNA samples. The researcher prepares a $5.0 \times 10^{-3} \text{ M}$ lithium acetate solution, which is then poured into the electrophoretic tank. After loading the samples into the agarose gel, a current is applied until the DNA fragments are separated (seen as individual bands in the gel). The agarose gel after electrophoresis is shown in Figure 2.

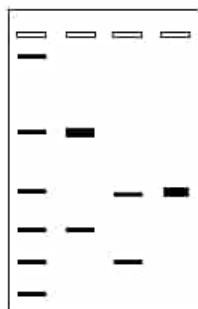


Figure 2 Agarose gel after electrophoresis using a $5.0 \times 10^{-3} \text{ M}$ lithium acetate solution (Note: The DNA fragments are represented as black bands.)

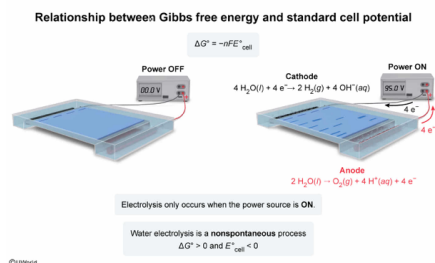
The electrolysis of water is a:

- ☐ A. spontaneous process with $\Delta G^\circ > 0$ and $E^\circ_{\text{cell}} < 0$. [1%]
- ☐ B. spontaneous process with $\Delta G^\circ < 0$ and $E^\circ_{\text{cell}} > 0$. [23%]
- ☒ C. nonspontaneous process with $\Delta G^\circ > 0$ and $E^\circ_{\text{cell}} < 0$. [71%]
- ☐ D. nonspontaneous process with $\Delta G^\circ < 0$ and $E^\circ_{\text{cell}} > 0$. [3%]

Correct

71% answered correctly

Explanation:



In an **electrolytic cell**, a **redox** reaction converts electrical energy (from an external source) into stored chemical energy. The **Gibbs free energy** ΔG° produced from this conversion represents the maximum amount of electrical work that can be performed during the reaction. The associated electromotive force (emf) is the amount of energy applied by the external power source per unit of electric charge. As such, the cell potential E°_{cell} for an electrolytic cell is equal to the emf, and ΔG° is related to E°_{cell} according to the relationship:

$$\Delta G^\circ = -nFE^\circ_{\text{cell}}$$

or

$$E^\circ_{\text{cell}} = \frac{-\Delta G^\circ}{nF}$$

where n is the number of electrons transferred per reaction and F is **Faraday's constant**. A redox reaction is **spontaneous** when $\Delta G^\circ < 0$ (ie, $E^\circ_{\text{cell}} > 0$) and **nonspontaneous** when $\Delta G^\circ > 0$ (ie, $E^\circ_{\text{cell}} < 0$).

For the experimental setup described in the passage, the electrolysis (ie, electrochemical **decomposition**) of water only occurs when the power supply is switched on. Because current from the power supply drives the electrolysis of water, the process is nonspontaneous. Therefore, $\Delta G^\circ > 0$ and $E^\circ_{\text{cell}} < 0$.

(Choices A, B, and D) The electrolysis of water is driven by an external power source, indicating a *nonspontaneous* reaction with $\Delta G^\circ > 0$ and $E^\circ_{\text{cell}} < 0$.

Educational objective:

The Gibbs free energy ΔG° for a redox reaction is related to the number of electrons transferred per reaction (n), the standard cell potential (E°_{cell}), and Faraday's constant (F), according to the equation $\Delta G^\circ = -nFE^\circ_{\text{cell}}$. For a spontaneous reaction, $\Delta G^\circ < 0$ and $E^\circ_{\text{cell}} > 0$; for a nonspontaneous reaction, $\Delta G^\circ > 0$ and $E^\circ_{\text{cell}} < 0$.

Time spent: 0 sec

Subject:
General Chemistry

QID: 403378

System:
Solutions and Electrochemistry